

SWIFT IDSS Sector Tables

Draft threshold, impact, and action framework for sector-specific space-weather decision support

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Core design principle: do not treat a NOAA scale level as a deterministic impact level. Use scale values as context, then evaluate regional/route/asset exposure, sector-specific indicators, and confidence to generate an IDSS action statement.

Purpose and caveats

This document turns the SWIFT sector tabs into draft five-level tables for hazard screening, possible impact, and IDSS action statements. It is meant as an application-design starting point, not as an official NOAA scale revision or sector-approved operating procedure.

The tables intentionally separate: (1) environmental hazard indicators, (2) sector exposure/vulnerability, (3) possible operational impact, and (4) recommended decision-support action. This matches the STPI finding that space-weather effects are localized and sector-specific, while the legacy scales are primarily phenomenon-based.

Thresholds are marked as either existing scale thresholds, proposed/working product thresholds, or partner-configurable thresholds. Values such as dB/dt, GIC amps, mission dose limits, CP excursions, and spacecraft charging limits should be configured with the relevant partner or operating authority before use in production.

Recommended SWIFT page pattern

Table A. Suggested SWIFT workflow

Step	UI function	Description
1	Select context	Sector, region/route/asset footprint, time window, and audience.
2	Evaluate hazard	Use the primary parameters in the sector table. Keep G/S/R/Kp visible as context, but do not force all impact logic through them.
3	Apply exposure modifiers	Account for geography, local time, system design, asset vulnerability, operating posture, and partner-specific thresholds.
4	Select impact level	Choose the level corresponding to plausible sector consequence if the hazard materializes.
5	Calculate risk posture	Combine probability and impact, then show confidence and primary driver.
6	Generate IDSS action	Display a short, actionable statement and a link to supporting products/plots/maps.

Universal risk matrix

Use the sector tables to select an impact level, then combine it with probability for the selected region/route/asset and time window. Probability bins should be configurable, but the starting point below is suitable for IDSS language.

Table B. Suggested probability labels

Probability label	Candidate bin	Plain-language meaning
Very Unlikely	<10%	Event/impact threshold is possible but not expected in the selected window.
Unlikely	10-30%	Low probability but meaningful enough to monitor for sensitive sectors.
As Likely as Not	30-60%	Reasonable chance; decision-makers should consider preparedness actions.

Probability label	Candidate bin	Plain-language meaning
Likely	60-80%	Expected or strongly favored; action statements should become more direct.
Very Likely	>80%	High-confidence occurrence or ongoing nowcast; execute appropriate procedures.

Table C. Suggested IDSS risk matrix

Impact level	Very Unlikely	Unlikely	As Likely as Not	Likely	Very Likely
5 Emergency / Severe	Moderate	High	High	Extreme	Extreme
4 Major	Minor	Moderate	High	High	Extreme
3 Moderate	Little / None	Minor	Moderate	Moderate	High
2 Minor	Little / None	Little / None	Minor	Minor	Moderate
1 Minimal	Little / None	Little / None	Little / None	Little / None	Minor

Common indicator dictionary

The sector tables reuse the following indicator families. The rows marked “prototype” should be treated as SWIFT design placeholders pending formal validation.

Table D. Indicator dictionary and threshold sources

Indicator family	Status	Candidate threshold logic	SWIFT use
NOAA G-scale / Kp	Existing NOAA scale	G1 Kp=5; G2 Kp=6; G3 Kp=7; G4 Kp=8; G5 Kp=9.	Useful global context for geomagnetic storming; not a direct local impact scale.
Hpo / regional K or H	Recommended direction / regionalization	Use if available for higher cadence, unbounded upper-end, and geographic specificity.	Supports upper-end storm differentiation and local/regional decision support.
Geoelectric field / E-field	Existing SWPC product plus proposed E-scale	E1 0.01-0.02; E2 0.02-0.05; E3 0.05-0.10; E4 0.10-0.20; E5 0.20-0.50; E6 0.50-1.00; E7 1-2; E8 2-5; E9 5-10; E10 10-20 V/km.	Primary grid/pipeline induction hazard indicator; SWIFT tables aggregate these to five levels.
dB/dt	Partner-configurable	Prototype bins used in some draft displays: <20, 20-40, 40-80, 80-120, >120 nT/min. Replace with local operating thresholds.	Useful local induction driver; must be interpreted with conductivity and asset geometry.
Measured/modelled GIC	Partner-configurable	Express as percent of utility/operator action threshold, not universal amps.	Best direct grid-response indicator where available.
NOAA S-scale	Existing NOAA scale	>=10 MeV proton flux: S1 10 pfu; S2 100; S3 1,000; S4 10,000; S5 100,000.	Good context for energetic proton environment; not a sufficient aviation dose proxy.
Higher-energy protons	Product/channel-specific	>30-50 MeV important for EVA/thin shielding; >100 MeV supports penetrating particle assessment; >=500 MeV is most relevant to aviation dose/GLE-style hard spectra.	Use energy spectrum and dose products rather than 10 MeV flux alone.
D-index / dose rate	Proposed aviation dose product	D0 <5; D1 5-10; D2 10-20; D3 20-40; D4 40-80; D5 80-160; D6 160-320; D7 320-640; D8 640-1280 uSv/h additional solar dose rate.	Primary aviation radiation-dose concept; assess route, altitude, latitude, and duration.
NOAA R-scale / x-ray class	Existing NOAA scale	R1 M1; R2 M5; R3 X1; R4 X10; R5 X20+ using GOES 0.1-0.8 nm x-ray flux.	HF absorption on sunlit side; not an all-frequency radio blackout scale.
D-RAP / absorption maps	Existing product family	Use regional absorption intensity and affected frequency/path rather than a single global number.	Primary HF operational map layer.
Scintillation S4 / sigma-phi / T-scale	Existing indices plus proposed product	Prototype S4 breakpoints: <0.3 insignificant; 0.3-0.4 weak; 0.4-0.6 moderate; >=0.6 strong. T-scale/CkL levels TBD.	Primary GNSS/SATCOM/radar ionospheric irregularity indicator.
TEC / TEC gradients / ROTI	Product/service-specific	Use local/regional thresholds tied to RTK/PPP/SBAS/timing performance.	Critical for GNSS precision, integrity, and timing applications.
Neutral density / drag	Prototype product need	Prototype percent change versus quiet/baseline: <10%, 10-25%, 25-50%, 50-100%, >100%.	Use for LEO drag, orbit prediction, SDA, and reentry uncertainty; no official NOAA scale.
Solar radio burst	Event/product-specific	Use receiver-band solar radio flux/noise and affected frequency bands when available.	Can affect GNSS, communications, and radar; not captured by R-scale.

Sector tables

Each table below is intended to be implemented as the sector-specific SWIFT “IDSS Sector Table.” The “Threshold indicators” column should support multiple simultaneous drivers; the page should show the primary driver rather than forcing a single index to control the whole sector.

1. Power Grid

Use this table for bulk-power, transmission, regional reliability, and utility-facing GMD IDSS. Kp/G-scale should remain visible as context, but the primary hazard logic should be regional geoelectric field, local dB/dt, and measured or modeled GIC response.

Primary parameters: regional geoelectric field (V/km), local dB/dt, measured/modelled GIC, regional K/Hpo, Kp/G-scale context, ground conductivity, transformer/grid topology, outage and reactive-reserve posture.

SWIFT table - Power Grid

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	E-field <0.05 V/km; dB/dt below local monitoring threshold, e.g., <20 nT/min prototype; measured/modelled GIC <10% of operator action threshold; Kp <5 / no G-level context.	No material operational impact expected. Routine GMD awareness only.	Continue normal monitoring. Keep geoelectric field, magnetometer, and GIC feeds visible for trend detection.	High confidence only where local magnetometer/E-field/GIC data are available.
2 Awareness	E-field 0.05-0.20 V/km; dB/dt rising or 20-40 nT/min prototype; GIC 10-25% of action threshold or weak response in susceptible assets; G1-G2 context possible.	Weak or localized GIC possible, especially in high-latitude, high-resistivity, long-line, or transformer-sensitive portions of the footprint.	Notify interested partners if trend increases. Verify data quality and watch local E-field/dB/dt/GIC, not Kp alone.	Use regional/utility thresholds where provided.
3 Elevated	E-field 0.20-1.0 V/km; dB/dt above monitoring threshold or 40-80 nT/min prototype; GIC 25-50% of action threshold or rapidly increasing; G2-G3 context.	Enhanced monitoring warranted. Localized alarms, transformer neutral current, VAR demand changes, or voltage-control concerns are plausible in susceptible systems.	Brief grid partners. Review current outages, maintenance, reactive reserve, and GMD operating procedure readiness.	Impact depends strongly on ground conductivity and actual grid posture.
4 High	E-field 1-5 V/km; dB/dt at/above operating threshold or 80-120 nT/min prototype; GIC 50-100% of action threshold; G3-G5/Hpo context.	Operational mitigation may be needed in affected regions. Regional voltage/reactive-power management and outage/maintenance posture may become relevant.	Coordinate with RC/BA/TOP partners. Review reactive reserve, defer vulnerable maintenance where appropriate, and prepare approved GMD procedures.	Do not infer impact from global G-scale alone.
5 Severe / Extreme	E-field >5 V/km, especially near or above the 8 V/km NERC reference level; dB/dt >120 nT/min prototype or extreme local derivative; measured/modelled GIC exceeds operator action threshold; G5/Hpo high-end context.	Emergency GMD posture possible. Reliability impacts are possible in susceptible systems; restoration and emergency-management coordination may be needed.	Execute approved GMD operating plan. Recall assets from outage where feasible, reconfigure to reduce GIC if procedurally allowed, coordinate restoration and emergency-management briefings.	Separate severe G5 from Carrington-class language unless supporting indicators justify it.

Source logic: STPI recommends geoelectric-field/GIC information for grid IDSS and notes that GIC response may not track G-scale cleanly. E-field bins aggregate the STPI suggested E-scale table into five SWIFT levels.

2. Aviation Operations

Use this table for dispatch, ATC coordination, polar-route planning, long-haul operations, alternate communications, reroute/fuel planning, and operational continuity. Keep this separate from the aviation radiation-dose table so communications/navigation decisions are not confused with dose decisions.

Primary parameters: route geometry, geomagnetic latitude, altitude, HF/D-RAP absorption, SATCOM availability, GNSS service quality, R-scale/x-ray flare, S-scale/proton/PCA context, G/Hpo context, flight phase, fuel and alternate routing constraints.

SWIFT table - Aviation Operations

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	No route-relevant HF absorption; R0; S0; G0-G1; SATCOM and GNSS nominal; no polar cap absorption along planned route.	Normal operations. Space weather does not require operational planning changes.	Display routine monitoring state. No special aviation IDSS action.	Route-specific view is essential; global scale levels alone are insufficient.
2 Awareness	G1-G2, S1-S2, or R1-R2 possible or occurring near route; weak HF absorption on sunlit side; weak polar absorption; high-latitude/polar route exposure exists but backup comms remain available.	Minor comms or navigation degradation possible. Operators may need increased awareness for polar or high-latitude routes.	Prompt dispatch/operations to monitor route-specific HF, SATCOM, GNSS, and radiation products. Confer with provider if G2/S2 or higher is expected.	STPI found some airlines confer with providers at G2/S2 and above, but procedures vary by company.
3 Elevated	G3, S3, or R3; moderate D-RAP/HF absorption affecting planned routing; SATCOM availability uncertain in polar region; GNSS degradation reports; proton/PCA indicators affecting high latitudes.	Route, altitude, or communications contingency may be needed. HF reliability may be reduced; SATCOM/GNSS availability should be checked.	Consider rerouting for SATCOM coverage, added fuel reserve for lower-altitude flight, alternate HF frequencies, and coordination with long-distance operational control/ANSPs.	Impact depends on planned route, aircraft equipage, time of day, and comms architecture.
4 High	G4, S4, or R4; strong HF absorption or polar cap absorption along route; SATCOM gap likely; GNSS integrity or timing effects plausible; radiation product approaching action threshold.	Major operational mitigation may be required for affected routes. Polar operations may require lower altitude, reroute, delay, or alternate communications.	Plan lower altitude or avoid vulnerable routes where procedures call for it. Coordinate dispatch, ATC/ANSP, provider, and management decision points.	Separate comms/navigation impacts from dose impacts.
5 Severe / Extreme	G5, S5, or R5; persistent route-relevant HF blackout/PCA; SATCOM/GNSS alternatives degraded; route-specific radiation product at high/action level; limited reliable communications over planned segment.	Operations on affected polar/high-latitude routes may be unacceptable without mitigation. Sustained reroute, descent, delay, or route avoidance may be needed.	Execute aviation space weather procedures. Actively reroute/descent/avoid affected region as required; brief crews and operational control; maintain contingency communications.	Avoid implying all global aviation is affected; scope to routes and equipage.

Source logic: STPI summarizes airline procedures using G/S/R levels to trigger conferencing, reroute/fuel planning, lower-altitude planning, and route changes when blackout areas affect planned routing.

3. Aviation Radiation Dose

Use this table only for human radiation exposure at aviation altitudes. It intentionally avoids using the NOAA S-scale alone as a dose proxy because the STPI report highlights that 10 MeV protons are not an adequate indicator for passenger/crew dose at commercial altitudes.

Primary parameters: dose-rate/D-index at route altitude and latitude, flight-specific accumulated dose, >500 MeV proton signatures, GLE/neutron monitor response, geomagnetic cutoff rigidity, altitude, route duration, and forecast confidence.

SWIFT table - Aviation Radiation Dose

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	D0: additional solar dose rate <5 uSv/h at route altitude/FL410 proxy; no GLE; no significant >=500 MeV proton enhancement; S-scale may be nonzero but dose product is quiet.	No aviation-dose action expected beyond routine radiation awareness.	Display dose product and route exposure. No reroute/altitude action for dose.	Use D-index/dose product as primary; S-scale is context only.
2 Awareness	D1-D2: 5-20 uSv/h additional solar dose rate; weak hard-spectrum or neutron-monitor indication; high-latitude/high-altitude route exposure.	Additional dose is elevated but likely manageable through monitoring and flight-specific assessment.	Monitor route-specific dose, update dispatch/crew advisories if required, and prepare mitigation options for polar/high-altitude operations.	Thresholds should be tied to airline, regulator, and radiation-protection procedures.
3 Elevated	D3-D4: 20-80 uSv/h; FAA-style 20 uSv/h alert concept may be met if sustained; >=500 MeV/GLE indicators support hard spectrum; flight-specific dose could become operationally relevant.	Operational concern for crew/passenger dose management on exposed routes. Altitude, latitude, and duration matter.	Run flight-specific dose assessment. Consider lower altitude, latitude change, route adjustment, or advisory products based on operator procedures.	This is the first level where dose-oriented IDSS should become prominent.
4 High	D5-D6: 80-320 uSv/h; strong hard-spectrum event; significant GLE/neutron monitor response; affected high-latitude/high-altitude routes.	Substantial additional dose possible for exposed flights. Mitigation may be warranted before or during flight.	Recommend route/altitude mitigation assessment. Coordinate operational control, radiation safety, provider, and management decision points.	Do not use S4 alone as the trigger; require dose or hard-spectrum confirmation.
5 Severe / Extreme	D7 or higher: >320 uSv/h; extreme hard-spectrum/GLE conditions; flight-specific dose potentially high over affected route segment.	Severe radiation-dose environment for exposed aviation routes. Avoidance or major mitigation may be needed.	Execute radiation-dose procedures: avoid affected high-altitude/high-latitude exposure, descend/reroute/delay as required, and document flight-specific dose.	Communicate clearly to avoid broad public alarm outside affected routes.

Source logic: STPI discusses a dose-rate-based D-index for aviation and notes that aviation dose concern is tied to higher-energy protons, not the S-scale 10 MeV threshold alone. D-index bins are from Meier and Matthiae 2014.

4. HF Communications

Use this table for HF skywave communications, including aviation, maritime, amateur radio, emergency backup HF, and polar HF. This page should be labeled HF Communications or X-ray HF Absorption rather than generic Radio Blackout.

Primary parameters: GOES x-ray class/R-scale, D-RAP absorption, sunlit-side footprint, frequency band, circuit geometry, polar cap absorption from energetic protons, S-scale/proton context, time of day, and backup comms availability.

SWIFT table - HF Communications

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	R0; x-ray below M1; D-RAP absorption minimal; no polar cap absorption on affected circuits.	HF propagation may vary normally, but no event-driven blackout expected.	Routine monitoring. Display sunlit-side and polar-route maps when available.	HF impacts are strongly frequency-, path-, and daylight-dependent.
2 Minor	R1 / M1-class threshold or weak D-RAP absorption; brief dayside degradation possible; weak PCA possible on high-latitude paths.	Weak or minor HF degradation on sunlit-side paths. Occasional loss of contact possible on vulnerable circuits.	Advise alternate frequencies and monitor absorption maps. Flag affected daylight hemisphere/path segments.	R-scale does not represent all radio frequencies.
3 Moderate	R2 / M5 threshold or moderate D-RAP absorption; limited blackout/fadeout on sunlit side; PCA may affect polar HF paths.	Moderate HF degradation. Some circuits may require frequency changes, alternate routing, or backup comms.	Issue sector-specific HF advisory. Recommend alternate frequencies/paths and backup comms check for affected operations.	Polar proton absorption can persist day/night, unlike x-ray dayside absorption.
4 High	R3 / X1 threshold or strong D-RAP absorption; significant dayside HF blackout for parts of the spectrum; S3+ proton/PCA context on polar paths.	Major HF disruption likely for affected paths. Operational users may lose reliable HF on selected circuits.	Activate backup communications. Coordinate with aviation/maritime/emergency users whose routes/AOR intersect the absorption region.	Tie alert to actual AOR, not just global R number.
5 Severe / Extreme	R4-R5 / X10 to X20+ or severe D-RAP absorption; prolonged or widespread dayside blackout; strong PCA affecting polar regions.	HF may be unusable for affected circuits. Critical operations relying on HF need non-HF alternatives.	Execute HF contingency procedures. Provide map-based timing, region, frequencies affected, and expected recovery updates.	Include restoration/recovery timing and confidence; avoid all-frequency "radio blackout" wording.

Source logic: NOAA R-scale is based on x-ray-driven HF interference on the sunlit side; STPI recommends clarifying that it is not a blackout across all radio frequencies.

5. Satellites / Spacecraft

Use this table for satellite owner/operators across LEO, MEO, GEO, HEO, and deep-space missions. The page should support anomaly attribution and operational posture, not imply a one-size-fits-all satellite impact.

Primary parameters: S-scale ≥ 10 MeV proton flux, higher-energy proton channels ($>50/>100/>500$ MeV as relevant), energetic electron flux and charging indicators, G/Hpo and regional geomagnetic context, neutral density/drag, orbit regime, shielding, mission mode, anomaly reports.

SWIFT table - Satellites / Spacecraft

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	S0; G0-G1; neutral density/drag near baseline, e.g., $<10\%$ prototype increase; energetic electron/proton levels below mission watch thresholds.	Routine spacecraft operations. No event-driven anomaly risk above normal background.	Normal monitoring and anomaly attribution logging.	Mission thresholds vary by orbit, shielding, design, and mode.
2 Awareness	S1 or G2; neutral density $+10\text{-}25\%$ prototype for LEO; elevated >10 MeV protons or charging environment below action threshold; electron flux rising.	Minor anomaly risk or drag awareness. Vulnerable payloads/components may need closer monitoring.	Increase environment monitoring. Flag sensitive maneuvers, uploads, imaging, or star-tracker operations for awareness.	For GEO, charging/electrons may dominate; for LEO, drag/density may dominate.
3 Elevated	S2-S3 or G3; elevated $>50/>100$ MeV proton channels; neutral density $+25\text{-}50\%$ prototype; charging indicators at/near mission watch threshold; LEO drag prediction errors increasing.	Anomaly risk elevated. SEUs, star tracker issues, charging, degraded imaging, or orbit prediction errors are plausible.	Brief spacecraft ops. Review safe-mode criteria, maneuver plans, payload schedules, conjunction screening, and anomaly attribution notes.	S-scale alone is insufficient; use mission environment channels.
4 High	S4 or G4; high >100 MeV proton channel or mission radiation threshold; neutral density $+50\text{-}100\%$ prototype; significant charging/radiation/drag conditions; anomaly reports emerging.	Significant operational risk for susceptible spacecraft. Attitude control, tracking, imaging, solar array, or command/telemetry impacts may require mitigation.	Consider delaying vulnerable operations, modifying payload activities, increasing tracking cadence, and coordinating SDA/conjunction assessment.	Thresholds should be configurable by mission class and orbit.
5 Severe / Extreme	S5 or G5/Hpo high-end; severe proton/electron environment; neutral density $>100\%$ prototype; major tracking/orbit-prediction degradation; multiple anomaly reports.	Severe environment for vulnerable assets. Loss of function, major drag/orbit uncertainty, or fleet-wide anomaly management possible.	Execute spacecraft storm procedures. Prioritize asset protection, tracking recovery, conjunction-risk reassessment, and executive/partner briefings.	Separate environmental severity from guaranteed spacecraft failure.

Source logic: STPI identifies satellite effects from energetic particles, charging, star tracker issues, and LEO drag/neutral-density concerns; it also notes stakeholder interest in neutral-density products.

6. GNSS / Navigation / Timing

Use this table for GNSS positioning, timing, integrity, augmentation, RTK/PPP, aviation navigation support, critical infrastructure timing, and radio-navigation users. The page should emphasize ionospheric conditions and receiver/service quality, not R-scale alone.

Primary parameters: scintillation S4 and sigma-phi, prototype T-scale/CkL if adopted, TEC/TEC gradients/ROTI, GNSS loss-of-lock/cycle slips, SBAS/GBAS/RTK/PPP correction quality, solar radio burst reports, G/Hpo context, receiver frequency/multi-constellation resilience.

SWIFT table - GNSS / Navigation / Timing

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	No regional scintillation product alert; S4 <0.3 prototype; sigma-phi below receiver concern; TEC gradients/ROTI normal; correction services nominal; no solar radio burst affecting GNSS bands.	GNSS accuracy, integrity, and timing expected to be normal.	Routine monitoring. Display regional ionosphere and service-quality indicators.	Use actual service-quality telemetry where available.
2 Awareness	Weak scintillation: S4 about 0.3-0.4 prototype or T1-T2; TEC gradients rising; G1-G2 context; occasional cycle slips or degraded precision reports.	Minor accuracy/confidence degradation possible for precision or single-frequency users.	Notify precision/timing users to monitor residuals, fix status, correction age, and timing holdover readiness.	Thresholds should be tuned to application: RTK, aviation integrity, timing, or public navigation.
3 Elevated	Moderate scintillation: S4 0.4-0.6 prototype or T3; sigma-phi elevated; TEC gradients disrupting corrections; G2-G3 context; regional loss-of-lock reports or correction-service degradation.	GNSS precision and timing may be degraded. RTK/PPP initialization delays, integrity alarms, or timing quality reductions are plausible.	Brief GNSS/timing partners. Recommend alternate timing sources, quality-control checks, and operational caution for precision tasks.	R-scale/x-ray flares are not the main GNSS ionosphere indicator; include solar radio burst separately.
4 High	Strong scintillation: S4 ≥0.6 prototype or T4; frequent cycle slips/loss of lock; severe TEC gradients; G4 context; augmentation/correction services degraded; solar radio burst affecting GNSS receiver bands.	Wide-area or regional degradation possible. Precision positioning and timing services may become unreliable in affected areas.	Activate alternate navigation/timing plans. Advise postponing precision-dependent operations where tolerances cannot be maintained.	Map the affected region/local time; equatorial and auroral regions differ.
5 Severe / Extreme	Severe/widespread scintillation or T5; persistent loss of lock; correction services unusable; timing users losing primary source; severe solar radio burst interference; G5/Hpo high context.	GNSS-dependent mission operations may be compromised in affected regions. Timing, navigation, and precision positioning may require full fallback.	Execute GNSS contingency procedures. Switch to backup timing/nav, suspend precision operations, and provide recovery timing updates.	Avoid global outage language unless telemetry supports it.

Source logic: STPI notes scintillation can degrade GNSS, SATCOM, and radar and that G/S/R scales only weakly connect to these effects. NOAA defines scintillation by S4 and sigma-phi; S4 numeric breakpoints here are prototype values commonly used for weak/strong scintillation.

7. Pipelines

Use this table for long conductive pipeline systems and cathodic-protection programs. This is related to the power-grid GMD hazard but the operational consequence is cathodic-protection disturbance and corrosion-risk management, not voltage stability.

Primary parameters: geoelectric field (V/km), local dB/dt, pipe orientation/length, ground conductivity, coating/CP system state, pipe-to-soil potential excursions, measured pipeline current if available, G/Hpo context.

SWIFT table - Pipelines

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	E-field <0.05 V/km; dB/dt below local monitoring threshold; CP readings normal; no measured pipeline current anomaly.	No event-driven CP impact expected.	Routine monitoring. No special pipeline IDSS action.	Pipeline response is highly asset-specific.
2 Awareness	E-field 0.05-0.20 V/km; dB/dt rising; weak CP noise or small pipe-to-soil potential excursions in susceptible regions.	Minor CP disturbance possible on long exposed segments.	Monitor CP telemetry and geoelectric field trends. Flag susceptible long pipelines and high-resistivity ground regions.	Use operator thresholds for CP compliance and integrity decisions.
3 Elevated	E-field 0.20-1.0 V/km; dB/dt above monitoring threshold; sustained CP excursions or modeled induced current in exposed segments.	Localized CP anomalies may require closer monitoring and post-event review.	Brief pipeline/integrity partners. Increase monitoring cadence and document CP deviations during event.	Do not map directly from Kp to CP impact.
4 High	E-field 1-5 V/km; dB/dt at operating threshold; significant CP excursions across multiple segments; high-risk orientation relative to E-field.	Widespread CP disturbance possible in exposed networks. Corrosion-risk management posture may be affected.	Coordinate with pipeline operators. Prioritize monitoring, defer sensitive CP surveys if needed, and plan post-event integrity review.	Geographic conductivity and asset orientation dominate risk.
5 Severe / Extreme	E-field >5 V/km, especially near/exceeding 8 V/km reference environment; severe dB/dt; major CP excursions or pipeline current above operator action threshold.	Severe induced-current environment. CP control and corrosion-risk management may require active response and post-event investigation.	Execute pipeline GMD procedures. Maintain safety/integrity monitoring and coordinate with emergency/grid partners where dependencies exist.	Use measured CP/current data where available.

Source logic: STPI and NOAA geoelectric field products connect induced currents in conductors to geomagnetic activity; thresholds should be validated with pipeline operators.

8. Human Spaceflight

Use this table for crewed spacecraft, EVA planning, lunar/deep-space trajectories, and mission operations. It should be tied to program flight rules and dosimetry, not generic public S-scale wording.

Primary parameters: >30-50 MeV proton environment, >100 MeV/hard-spectrum indicators, onboard dosimeters and modeled dose behind shielding, S-scale context, EVA/suit exposure, storm shelter capability, orbit/trajectory/geomagnetic shielding, mission phase.

SWIFT table - Human Spaceflight

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	No SEP concern; >30-50 MeV proton channels quiet; onboard/model dose rates normal; no EVA/timeline sensitivity.	Normal mission timeline and EVA planning.	Routine radiation monitoring. No special crew action.	Program-specific dose limits and flight rules control decisions.
2 Awareness	S1 or rising SEP watch; weak >30 MeV channel increase; dose below program watch threshold; EVA/thin-shielded activity possible within forecast window.	Dose environment requires attention but no immediate crew action expected.	Increase radiation monitoring. Review EVA/timeline exposure and storm-shelter readiness.	S-scale is context; crew dose depends on energy spectrum and shielding.
3 Elevated	S2-S3 or significant >30-50 MeV proton flux; modeled/onboard dose approaching program watch threshold; EVA or lightly shielded activity scheduled.	Crewed operations may need timeline review. EVA deferral or additional shielding posture may be considered.	Brief flight control and radiation safety. Reassess EVA, trajectory exposure, and shelter timelines.	STPI notes astronauts can receive increased dose from 30-50 MeV protons in thin shielding.
4 High	S4 or strong >30-100 MeV event; dose rate reaches program action threshold; hard spectrum or GLE indication; EVA in progress or planned.	Active mitigation may be required. EVA restrictions, termination, or shelter posture may be needed.	Execute flight-rule-driven actions: delay/terminate EVA, move crew to better shielded area, update mission timeline.	Use real-time dosimetry when available.
5 Severe / Extreme	S5 or extreme hard-spectrum SEP/GLE; high onboard/model dose; shelter thresholds exceeded or likely; limited geomagnetic shielding/mission-critical exposure.	Crew health-critical radiation posture possible. Major mission timeline changes may be required.	Execute crew radiation emergency procedures. Shelter, abort EVA, protect critical crew tasks, and coordinate mission leadership.	Avoid public/aviation assumptions; this is mission-architecture-specific.

Source logic: STPI states aviation dose concern is tied to >=500 MeV protons, while astronauts can be affected by roughly 30-50 MeV protons in spacesuits or thin shielding.

9. Emergency Management / Critical Comms

Use this table for emergency managers, public safety, critical communications, and continuity-of-operations partners. It should combine space weather hazards into a practical question: which communications, navigation/timing, and critical infrastructure dependencies remain reliable in the affected area?

Primary parameters: HF/R-scale/D-RAP, GNSS/scintillation/TEC, SATCOM impacts, power-grid GMD risk, regional E-field/GIC products, critical infrastructure dependencies, emergency operations posture, public information needs.

SWIFT table - Emergency Management / Critical Comms

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	No route/AOR-relevant HF, GNSS, SATCOM, or grid risk; R0; G0-G1; no regional scintillation alert; E-field quiet.	No space-weather-driven emergency-management action expected.	Routine monitoring. Keep dashboard available for situational awareness.	Scope to the emergency manager AOR, not global conditions.
2 Awareness	R1-R2 or weak HF absorption; G1-G2; weak scintillation indicators; grid/pipeline E-field awareness level; no critical service degradation reported.	Minor degradation possible. Backup systems should remain available.	Issue awareness note if partner requested. Recommend checking backup comms, timing holdover, and utility status.	Avoid alarmist language for public-facing comms.
3 Elevated	R2-R3 or route/AOR HF degradation; G3; moderate scintillation/TEC gradients; E-field elevated; localized grid/comms partner concerns.	Emergency communications or timing may be degraded regionally. Some backup pathways may be needed.	Brief emergency managers. Identify affected frequencies/services, alternate comms, timing backups, utility points of contact, and public-message needs.	This level should generate plain-language action statements.
4 High	R4 or major D-RAP/PCA; G4 or high regional E-field; strong scintillation/GNSS degradation; SATCOM issues; grid risk high in AOR.	Multiple communications or infrastructure dependencies may need active management.	Activate comms contingency planning. Coordinate with utilities, telecoms, aviation/maritime/public safety, and state/federal partners.	Include confidence and expected duration.
5 Severe / Extreme	R5 or severe/Persistent HF/PCA; G5/Hpo high; severe GNSS/timing disruption; E-field severe; critical infrastructure partners report operational impacts.	Critical comms, timing, or infrastructure dependencies may be unreliable in affected regions.	Execute emergency coordination posture. Provide regular IDSS briefings, fallback comms guidance, utility status integration, and public-facing risk messaging.	Do not imply nationwide failure unless partner data support it.

Source logic: STPI recommends sector-specific risk-informed action statements and notes emergency managers need information about reliable communication systems in their disaster zones.

10. Public / Aurora

Use this table for public aurora visibility and public-facing space weather awareness. It should generally not be framed as public safety risk unless other sector tables indicate infrastructure or communications risk.

Primary parameters: Kp/G-scale and regional auroral oval/viewline, geomagnetic latitude, local night/clouds/light pollution, Hpo/high-end context, public interest, and any separate infrastructure risk statements from grid/comms sectors.

SWIFT table - Public / Aurora

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	Kp <4 or aurora confined to usual high latitudes; regional aurora forecast quiet; no public infrastructure message needed.	Normal public conditions. Aurora unlikely outside high-latitude areas.	No special public action. Provide routine educational content if desired.	Weather/clouds affect visibility but are not space weather indicators.
2 Minor / Interest	Kp 4-5 / G1 context; aurora possible at high latitudes or northern-tier viewing areas; public interest rising.	Aurora viewing possible for high-latitude/northern users. Minimal public-safety implications.	Post aurora visibility guidance and explain that visibility does not equal infrastructure impact.	Use regional auroral oval/viewline instead of Kp alone.
3 Moderate / Broad Interest	Kp 6 / G2 context; aurora possible farther equatorward, around typical G2 visibility guidance; social/media interest likely.	Broader aurora viewing opportunity. Some users may ask about technology impacts.	Provide public explainer with separate links to operational impacts if needed. Coordinate with media/social teams.	Keep public language simple and non-alarmist.
4 High / Widespread Visibility	Kp 7-8 / G3-G4 context; aurora possible into mid-latitudes; high public interest; separate sector products may show GNSS/HF/grid impacts.	Widespread public attention and possible misinformation risk. Aurora may be highly visible where skies cooperate.	Issue coordinated public messaging: what to expect, where aurora may be visible, and what impacts are/are not expected.	Use separate risk sections for grid/comms, not aurora color alone.
5 Exceptional / Historic Visibility	Kp 9 / G5 or high Hpo context; aurora possible unusually far equatorward; potentially historic event; separate IDSS sectors should assess infrastructure risk.	Exceptional public attention. Public may conflate aurora with catastrophic infrastructure impacts.	Provide frequent plain-language updates and clear separation between aurora visibility, space weather conditions, and any actual infrastructure advisories.	Avoid "extreme means catastrophic" unless sector data support it.

Source logic: NOAA G-scale includes aurora visibility examples by geomagnetic latitude, while STPI recommends differentiating severe/non-catastrophic events from truly catastrophic events.

11. Space Domain Awareness

Use this table for catalog maintenance, tracking, conjunction assessment, reentry prediction, LEO operator coordination, and space traffic coordination. It is related to the satellite table but focuses on orbit knowledge and tracking uncertainty rather than spacecraft health.

Primary parameters: modeled/observed neutral density, drag acceleration, percent density/drag increase versus quiet/baseline, G/Hpo/Ap context, LEO altitude and ballistic coefficient, tracking residuals, covariance growth, conjunction screening workload, reentry uncertainty.

SWIFT table - Space Domain Awareness

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	Neutral density/drag near baseline, e.g., <10% prototype increase; G0-G1; tracking residuals and covariance nominal.	Normal orbit determination and conjunction assessment quality.	Routine tracking, screening, and catalog maintenance.	No official neutral-density scale currently exists; percent thresholds are prototype.
2 Awareness	Neutral density +10-25% prototype; G1-G2; early tracking residual increase for very low LEO or high area-to-mass objects.	Minor drag awareness. OD updates may be needed for sensitive LEO objects.	Increase monitoring of low-perigee and high-drag objects. Flag likely covariance growth.	Tune thresholds by altitude, ballistic coefficient, and tracking cadence.
3 Elevated	Neutral density +25-50% prototype; G3; drag model residuals increasing; conjunction screening sensitivity affected in LEO.	Orbit prediction uncertainty is elevated. Conjunction assessments and maneuver planning may require more frequent updates.	Increase tracking/OD cadence. Re-run conjunction screenings and notify affected LEO operators.	G-scale is a proxy; use density/drag products when available.
4 High	Neutral density +50-100% prototype; G4; significant residuals/covariance growth; catalog maintenance burden high; reentry predictions shifting.	Large orbit-prediction uncertainty for affected LEO population. Collision-risk management may be degraded.	Coordinate SDA operators. Prioritize tracking, update reentry/conjunction products, and communicate uncertainty bounds.	Communicate uncertainty explicitly.
5 Severe / Extreme	Neutral density >100% prototype; G5/Hpo high-end; severe drag/residuals; major catalog disruption or widespread LEO covariance growth.	Severe tracking/orbit-prediction degradation. Conjunction assessment and reentry prediction may be unreliable until recovery.	Execute SDA storm posture. Prioritize critical objects, communicate catalog uncertainty, coordinate with satellite operators and national/international partners.	STPI notes neutral-density products are an evolving research area.

Source logic: STPI identifies neutral density/upper-atmosphere expansion as a key satellite/SDA need and notes that users suggested percent-change products.

12. Surveying / Geomatics

Use this table for magnetic surveys, GNSS surveying, RTK/PPP, photogrammetry ground control, precision agriculture-style positioning support if needed, and geodetic operations. It should distinguish magnetic-field disturbance from GNSS ionospheric disturbance.

Primary parameters: local K/Kp/regional Hpo, dB/dt, magnetometer noise, GNSS scintillation S4/sigma-phi, TEC gradients/ROTI, correction service status, receiver fix/float state, baseline length, survey tolerance and QA/QC flags.

SWIFT table - Surveying / Geomatics

Level	Threshold indicators to evaluate	Operational impact statement	Suggested IDSS action statement	Confidence / implementation notes
1 Routine	Local K <4; dB/dt quiet; S4 <0.3 prototype; TEC gradients/ROTI normal; RTK/PPP/correction service nominal; magnetic survey residuals normal.	Precision surveying and magnetic operations expected to perform normally.	Routine QA/QC and normal field operations.	Use project tolerance as the final impact threshold.
2 Awareness	Local K 4-5 or G1 context; dB/dt rising; weak scintillation S4 0.3-0.4 prototype; correction age/residuals slightly degraded.	Minor delays or QA/QC flags possible. Sensitive magnetic surveys or long-baseline GNSS may need attention.	Advise crews to monitor fix status, residuals, PDOP/GDOP, correction age, and magnetometer stability.	Effects vary with receiver, baseline, constellation, and local ionosphere.
3 Elevated	Local K 6 or G2 context; moderate scintillation S4 0.4-0.6 prototype; TEC gradients disrupt RTK/PPP; magnetic noise exceeds project watch threshold.	RTK initialization delays, float solutions, lower confidence, or magnetic survey noise likely for sensitive work.	Consider schedule adjustments for high-precision tasks. Increase check shots, independent controls, and post-processing review.	Flag in pre-survey report and client risk notes.
4 High	Local K 7-8 or G3-G4 context; strong scintillation S4 >=0.6 prototype; frequent cycle slips/loss of lock; dB/dt/magnetic noise above operating threshold.	High-precision GNSS or magnetic survey operations may be unreliable in affected region.	Postpone sensitive magnetic/GNSS work if tolerances cannot be maintained. Use alternatives or additional controls.	Scope by region and local time; equatorial/auroral effects differ.
5 Severe / Extreme	Local K 9/G5 or high Hpo; severe/widespread scintillation; persistent loss of lock; correction services unusable; magnetic disturbance severe.	Precision operations likely unreliable. Significant rework risk if operations continue without mitigation.	Suspend or shift precision operations; move to redundant methods; document event for QA/QC and client communication.	STPI had limited direct engagement with some adjacent sectors, so validate with geomatics users.

Source logic: STPI includes surveying among user sectors and notes operational users want geographic specificity. GNSS thresholds follow the GNSS/scintillation logic above and should be validated with survey stakeholders.

Implementation notes for SWIFT

Table E. Suggested implementation requirements

SWIFT component	Recommendation
Header card	Sector, region/route/asset, time window, current risk posture, probability, confidence, primary driver, and action statement.
Threshold engine	Allow more than one threshold family to contribute. Store the trigger source, value, unit, time window, and product timestamp.
Partner configuration	Support partner-specific thresholds for GIC amps, dB/dt, CP excursions, dose action levels, satellite charging, mission radiation, and GNSS performance.
Confidence	Show confidence separately from probability. Examples: model-only, nowcast-confirmed, local sensor-confirmed, partner-reported impact.
Geographic specificity	Require a footprint: utility area, pipeline segment, flight route, satellite orbit shell, GNSS service region, or emergency-management AOR.
Action statements	Use verbs that match IDSS: monitor, brief, confer, review, prepare, coordinate, execute. Avoid deterministic language unless observed impacts confirm it.
Public messaging	Keep public aurora communication separate from operational risk pages. Aurora visibility does not automatically imply infrastructure consequences.
Validation	Before production, validate each sector table with representative users and collect false alarm/missed event feedback.

Reference notes

STPI / IDA. Revisions to NOAA's Space Weather Scales. Final report, IDA Product 3003755, 2025.

NOAA SWPC. NOAA Space Weather Scales. Current web guidance for G-, S-, and R-scale definitions.

NOAA SWPC. Geoelectric Field Models (US Canada 1D & 3D EMTF CONUS). Current product page and usage notes.

NOAA SWPC. Ionospheric Scintillation. Product/phenomenology page describing S4 and sigma-phi indices.

Meier, M. M., and D. Matthiae. 2014. A Space Weather Index for the Radiation Field at Aviation Altitudes. Journal of Space Weather and Space Climate 4:A13.

Australian Bureau of Meteorology / SWS. About Ionospheric Scintillation. Used only for prototype S4 interpretation breakpoints.

NERC TPL-007 references are used only as context for the 8 V/km benchmark mentioned in the STPI report; operator-specific thresholds should control operational actions.